PARAKH — Stage 6 PRD

Confidence-Gated Decision & Routing Engine

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1. Objective

Convert the pair:

$$
(R(r), C(r))
$$

into a **clear, auditable decision** for each record.

This stage defines the **final system output**:

- INVESTIGATE
- REMEDIATE
- MONITOR
- CLEAR

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2. Core Principle

> **Do not act on data you cannot trust.**

A high-risk signal on low-confidence data is **not a fraud signal**—it is a **data quality problem**.

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3. Scope

Included

- Decision logic using (R) and (C)
- Threshold definition
- Routing categories
- Decision explanations
- Output formatting

Excluded

- Confidence computation (Stage 2)
- Risk computation (Stage 5)
- UI (Stage 7)

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4. Inputs

From Stage 2:

```
confidence_score: float  # C(r)
```

From Stage 5:

```
risk_score: float  # R(r)
```

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5. Output

Each record must produce:

```
{
  "decision": str,  # INVESTIGATE | REMEDIATE | MONITOR | CLEAR
  "confidence": float,
  "risk": float,
  "reason": str
}
```

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6. Decision Logic

6.1 Thresholds

Define:

```
theta_C = 0.6   # confidence threshold
theta_R = 0.7   # risk threshold
```

6.2 Routing Function

$$
\text{route}(r) =
\begin{cases}
\text{INVESTIGATE} & R \ge \theta_R \;\wedge\; C \ge \theta_C \\
\text{REMEDIATE} & C < \theta_C \\
\text{MONITOR} & R \ge \theta_R \;\wedge\; C < \theta_C \\
\text{CLEAR} & \text{otherwise}
\end{cases}
$$

\end{cases}
$$

6.3 Priority Rule (IMPORTANT)

Apply rules in order:

```
if C < theta_C:
    decision = "REMEDIATE"
elif R >= theta_R and C >= theta_C:
    decision = "INVESTIGATE"
elif R >= theta_R:
    decision = "MONITOR"
else:
    decision = "CLEAR"
```

> Ensures low-confidence always routes to remediation first.

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7. Decision Semantics

INVESTIGATE

- High risk
- High confidence
  → **credible anomaly**

REMEDIATE

- Low confidence
  → **data unreliable**

MONITOR

- High risk but low confidence
  → **potential issue after data fix**

CLEAR

- Low risk
  → **no action required**

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8. Explanation Generation (CRITICAL FOR DEMO)

Each decision must include a human-readable reason.

8.1 Format

```
reason = f"""
Record classified as {decision} because:
- Confidence = {C:.2f}
- Risk = {R:.2f}
- Key signals:
  - Cost anomaly: {z_cost}
  - Vendor concentration: {hhi}
  - Burst activity: {burst}
  - Duplicate similarity: {dup}
"""
```

8.2 Rules

- Always include:
    - top contributing signals
- Keep explanation < 5 lines
- Must be understandable by non-technical user

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9. Implementation Architecture

9.1 Modules

- `router.py`
- `decision_engine.py`
- `explain.py`

9.2 Core Class

```
class Router:
    def __init__(self, theta_r=0.7, theta_c=0.6):
        self.theta_r = theta_r
        self.theta_c = theta_c

    def route(self, risk_scores, confidence_scores):
        return DecisionResult
```

9.3 Output Object

```
class DecisionResult:
    decisions: List[str]
    explanations: List[str]
```

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10. Non-Functional Requirements

Determinism

- Same input → same decision

Interpretability

- Every decision must be explainable

Performance

- < 1 second for 50k records

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11. Validation & Testing

11.1 Rule Testing

Test all cases:

C	R	Expected
High	High	INVESTIGATE
Low	High	REMEDIATE
High	Low	CLEAR
Low	Low	REMEDIATE

11.2 Boundary Testing

- C = theta_C
- R = theta_R

Ensure correct classification

11.3 Distribution Check

```
count(decisions)
```

Ensure:

- Not all records in one category

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12. Edge Cases (MANDATORY)

Handle:

- Missing R or C → assign REMEDIATE
- All records low confidence
- All records high risk
- Uniform scores

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13. Acceptance Criteria

Stage complete if:

- [ ] Every record assigned a decision
- [ ] Explanations generated
- [ ] Threshold logic works correctly
- [ ] Edge cases handled safely
- [ ] Output interpretable

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14. Definition of Done

- Full decision pipeline operational
- Ready for final presentation layer (Stage 7)

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15. Key Design Principle

> **Separate uncertainty from suspicion.**

- Confidence = "Can we trust this data?"
- Risk = "Is this behavior unusual?"

The system acts only when **both are high**.
